

Retail Sales & Customer Analytics Using Tableau

Prepared by: Haathim Manaf

Project Type: Business Intelligence & Data Visualization

Tool: Tableau

1. Executive Summary

This project analyzes retail sales performance and customer behavior using Tableau. Two interactive dashboards were developed to monitor business KPIs, evaluate product profitability, and understand customer purchasing patterns.

The objective is to provide actionable insights for sales growth, inventory optimization, and customer retention strategies.

2. Business Objectives

The analysis was conducted to:

- Monitor total sales, profit, and quantity trends
 - Identify profitable and loss-making product subcategories
 - Analyze weekly sales and profit fluctuations
 - Understand customer order behavior and distribution
 - Identify top customers contributing to revenue
 - Support data-driven marketing and inventory decisions
-

3. Data Model & Architecture

A dimensional data model was implemented for analytics.

Fact Table

- **fact_orders**
 - Order-level transactional data including sales, profit, quantity, order date, customer ID, and product ID.

Dimension Tables

- **dim_customers** – customer attributes
- **dim_products** – product category and subcategory information
- **dim_location** – geographical attributes

This structure supports scalable filtering and multi-dimensional analysis.

4. Key Performance Metrics

Sales Dashboard KPIs

- **Total Sales:** \$733K
- **Total Profit:** \$93K
- **Total Quantity Sold:** 12K units
- **Year-over-Year Growth:**
 - Sales: +20.4%
 - Profit: +12.5%
 - Quantity: +26.8%

Customer Dashboard KPIs

- **Total Customers:** 693 (+8.6% YoY)
 - **Average Sales per Customer:** \$1,058 (+10.8% YoY)
 - **Total Orders:** 1,687 (+28.3% YoY)
-

5. Sales & Product Performance Analysis

Subcategory Performance

- Subcategories such as Phones, Chairs, and Binders contribute significantly to total sales.
- Some subcategories show negative profit contribution, indicating pricing or cost inefficiencies.

Weekly Sales & Profit Trends

- Sales and profit exhibit seasonal peaks and troughs across the year.
- Several weeks fall below average profit thresholds, indicating demand volatility.

Business Implication

- High-performing products should be prioritized in inventory and promotions.
 - Loss-making subcategories require pricing optimization or cost reduction strategies.
-

6. Customer Analytics Insights

Customer Distribution by Order Frequency

- Majority of customers place **1–3 orders**.
- Very few customers place more than 5 orders, indicating low repeat purchase rates.

Top Customers by Profit

- A small group of customers contributes a large share of total profit.
- High-value customers show repeat purchase behavior and higher sales volumes.

Business Implication

- Customer retention strategies should focus on high-value customers.
 - Low-frequency customers can be targeted with personalized marketing campaigns.
-

7. Key Business Insights

- Revenue growth is driven mainly by increased order volume rather than customer acquisition alone.
 - Profitability varies significantly across product subcategories, highlighting portfolio optimization opportunities.
 - Customer revenue follows a Pareto pattern where a small segment contributes a large share of profit.
 - Seasonal demand patterns impact weekly sales and profit fluctuations.
-

8. Strategic Recommendations

Customer Strategy

- Implement loyalty programs for high-value customers.
- Use targeted promotions to increase repeat purchases among low-frequency customers.

Product Strategy

- Optimize pricing or discontinue consistently loss-making products.
- Focus marketing and inventory on high-margin subcategories.

Sales & Marketing Strategy

- Align campaigns with seasonal demand peaks.
- Use weekly trend insights for discount planning and stock replenishment.

BI & Analytics Strategy

- Integrate dashboards with automated data pipelines for real-time monitoring.
- Extend analysis with customer segmentation and cohort analysis.

9. Tools & Skills Demonstrated

- Tableau Dashboard Design
- Calculated Fields & KPI Indicators
- Data Modeling (Fact & Dimension Tables)
- Retail Sales & Customer Analytics
- Data Visualization & Business Storytelling

10. Deliverables

- Two interactive Tableau dashboards (Sales & Customer)
- Structured analytical report (this document)
- Project design mockups and implementation phases
- Public Tableau dashboard link
- GitHub repository with datasets and documentation

Tableau Public Link

https://public.tableau.com/views/SalesCustomerDashboards_17692811875880/CustomerDashboard